

Xiao XUE

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RESEARCH INTERESTS

Human-Computer Interaction (HCI), Responsible AI, Participatory Design, Research through Design

EDUCATION

The University of Edinburgh, UK - MSc Design Informatics

09/2020 – 08/2021

- Degree classification: **Distinction**; Advisor: **John Vines**

Relevant Modules: Analyzing Qualitative Data, The Human Factor: Working with Users, Histories and Future of Technology, Data Science for Design, Design with Data

The Hong Kong Polytechnic University - BSc (Hons) Internet & Multimedia Technologies

09/2016 – 08/2020

- Dean's Honours List (2019/ 2020) in Faculty of Engineering; Best Semester GPA Award (2019/ 2020 Semester 1,2 and 3) in the EIE Department

Relevant Modules: Object-Oriented Programming, Computer Vision and Pattern Recognition, Multimodal Human-Computer Interaction Technology, Computer Game Development

PUBLICATIONS

- Designing with Tensions: Reframing Smart Home Ecosystem UX through In-the-Wild Evidence
He, S., Liu, Y., Wang, Z., Xue, X., Zhang, X., Shi, L., & Fu, X. CHI EA 2026
- Hear Us, then Protect Us: Navigating Deepfake Scams and Safeguard Interventions with Older Adults through Participatory Design
Zhai, Y., Xue, X., Guo, Z., Jin, T., Diao, Y., & Jeung, J. CHI 2025.
- When My Car Whispers Secrets to My Home: Envisioning the Opportunities and Challenges of Human-Vehicle-Home Integrated Ecosystems through Probes
Xue, X., He, S., Chen, K., Qiao, R., Xu, W., Wang, E., Zhao, Y., & Fu, X. CHI EA 2025.
- A Card-Based Co-Design Toolkit for Exploring Smart Material Applications with Multiple Stakeholders: A Case Study on Automotive Interior Design
Yu, T., Wang, X., Lu, Y., Yu, K., Yao, X., Deng, W., Li, Z., Li, X., Xue, X., Yang, Y., Guo, Y., Yao, Y., Liu, G., & Mi, H. DIS 2025.
- Who Should Hold Control? Rethinking Empowerment in Home Automation among Cohabitants through the Lens of Co-Design
Xue, X., Li, X., Jia, B., Du, J., & Fu, X. CHI 2024.

Manuscripts Under Review

- Beyond Driving: Envisioning Activities in Future Autonomous Vehicles through Experience-Centered Design
Chen, K., Xue, X., Tan, R., & Lyu, C. Under revision (Travel Behaviour and Society).

RESEARCH PROJECTS

Only-for-Workers AI Tools: A Speculative Co-Design Workshop on Workplace AI

2026

- Ongoing project using the provocative prompt of AI tools that “only help workers, not bosses” within a Magic Machines co-design workshop to examine how knowledge workers imagine genuinely supportive AI amid surveillance, datafication, performance management, replacement anxiety, and organizational power in workplace AI deployment.
- Analyzing workers' concepts and boundary discussions to identify what workplace AI risks can be meaningfully addressed through system-level design implications, and what responsibilities may exceed the scope of HCI and AI design.

Investigating Generative AI Use in High-School Computer-supported Collaborative Learning

2026

- Study examining how generative AI functions as a shared but unevenly accessible group resource in collaborative learning, focusing on participation, collective envisioning, and potential marginalization in AI-mediated group work.
- Contributed to the methodological design of an AI-mediated design sprint and follow-up co-design activity with high-school students from Beijing and Tibet; conducted reflexive thematic analysis of observations, artifacts, worksheets, and interviews; and supported manuscript writing and refinement.

How and Why Non-AI Experts Enact Design-in-Use with Commercial LLMs in the Wild: Meaning-Making, Agency, and Contestability

2025

- First-author study examining how non-AI experts make sense of, adapt, and contest commercial LLMs through everyday design-in-use practices.
- Designed and conducted participatory semi-structured interviews with 12 lead users, organized reflexive thematic analysis, and led manuscript writing and contribution positioning around agency, participation, and platform-governed AI.

Envisioning Human–Vehicle–Home (HVH) Integrated Ecosystems with People through Probes **2025**

- Led a user-centered probe study on car–home integration in China, directing probe design, recruitment, interviews, thematic analysis, and manuscript writing to develop design implications around cross-space well-being, family support, privacy, safety, and shared use.

Research on Safeguarding Older Adults from Deepfake Scam **2024**

- Designed and facilitated participatory workshops with 10 older adults, using simulated deepfake scam videos and provocative safeguarding designs to examine their risk perceptions and protection expectations.
- Contributed to reflexive thematic analysis, highlighting autonomy, digital literacy, and shared responsibility as key considerations for tailored deepfake-scam safeguards.

Rethinking Empowerment in Home Automation among Cohabitants through the Lens of Co-Design **2023**

- First-author study rethinking empowerment in home automation by examining how cohabitants negotiate control, autonomy, and participation when constructing Trigger-Action Program schemes together.
- Designed card-based tools and conducted workshops with cohabitants, leading content/thematic analysis and manuscript writing to reveal how automation can both marginalize residents and create opportunities for perceptible, participatory, and contestable control.

WORK & RESEARCH EXPERIENCE

Research Intern – The Future Laboratory, Tsinghua University **10/2022–Present**

Mentored by Dr. Xinyi Fu

- Contributed to HCI/design research, academic writing, grant preparation, and scholarly service, including living-lab coordination, demonstrations, academic/industry visits, and research community engagement.
- Led and contributed to five CHI/DIS publications, taking major responsibility for study design, participant engagement, qualitative and design-led analysis, manuscript development, and scholarly argumentation.

Research Assistant – Nanyang Technological University **05/2019–08/2019**

Mentored by Prof. Perry Ping Shum

- Worked with computer vision researchers on an emotion recognition system and AR presentation host for an interactive Xperience Room environment.

ACADEMIC SERVICE

Reviewer: CHI 2025–2026; Foundations of Digital Games (FDG) 2026; GamiFIN 2022–2023; Chinese CHI 2023

Program Committee: Chinese CHI 2024

EXTRA-CURRICULAR ACTIVITIES & VOLUNTARY WORK

- Dr. Y.K. Ching Memorial Scholarship, 2017/18.
- External Vice President, Chinese Mainland Students' Association, PolyU, 2017–2018; sourced over HKD 300,000 in sponsorship and organized an inter-university singing competition with 700+ attendees.
- Volunteer teacher, HeartFire 17th voluntary teaching trip, 2018; completed 200 hours of teaching and activity facilitation for 100+ students in Hunan, China.

SKILLS

Qualitative methods: thematic analysis, content analysis, reflexive thematic analysis, Research through Design (participatory design and speculative design methods)

Quantitative methods: survey design, statistical analysis, data visualization.

Technical skills: Python, Java, C/C++, SQL, PHP; Unity prototyping; Adobe CC; 3ds Max.

Languages: Mandarin Chinese (native), English (professional), Cantonese (intermediate).